

Principal components and singularities of orthogonal frame varieties

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Abstract

We study the defining ideals and singularities of varieties of pairwise orthogonal vectors. For $q = \binom{k}{2}$ and $h = q + k$, we determine the ideal of the principal component of the variety of h orthogonal vectors in dimension $2q$ over $\mathbb{C}$. Besides the Gram quadrics, the ideal has generators in the single degree $q(k+1)$, constructed by Gale duality from a Veronese determinant and indexed by two orthogonal-group representations. In the first case, six vectors in dimension six, this gives fifteen quadrics and 330 explicit equations of degree twelve. In every characteristic different from two, the principal components in this family are normal, satisfy S_k but not S_{k+1} , and have depth one below their dimension. We determine their deficiency modules and regularity. We also construct normal, and frequently factorial, orthogonal frame varieties at the prime threshold that fail to have rational singularities in characteristic zero and fail to be F -rational in every odd characteristic. The corresponding numbers of columns form a set of natural density one half. An exact local chart identifies the generic transverse singularities with cones over smooth complete intersections of quadrics and gives their classification. These results answer the first defining-ideal case and the rationality question posed by Casabella and Sammartano, and give normal but non-Cohen–Macaulay principal components.

1 Introduction

Let E be a nondegenerate quadratic space of dimension M over an algebraically closed field K of characteristic different from two. For a generic matrix $V = [v_1 \ \cdots \ v_h] \in \text{Mat}_{M,h}$, set

$$S = K[v_{ai}], \quad \mathfrak{l}(M, h) = (\langle v_i, v_j \rangle : 1 \leq i < j \leq h), \quad X(M, h) = \text{Spec } S/\mathfrak{l}(M, h).$$

When $M \geq h$, the closure of the locus on which all column norms are nonzero is an irreducible component. We denote it by $C(M, h)$ and call it the *principal component*; its prime ideal is $P(M, h)$.

Casabella and Sammartano determine the irreducible components of orthogonal frame varieties and the thresholds for complete intersection, primality, and normality [2]. Their Questions 8.1, 8.3, and 8.4 ask, respectively, about rational singularities at the prime threshold, defining equations of the principal component, and its normality and Cohen–Macaulayness. The first unresolved defining-ideal case identified there is $(M, h) = (6, 6)$.

Our results concern two families adjacent to these thresholds. The first lies at a complete-intersection threshold where the full scheme has three components. The second lies at a prime threshold. Both are governed by the matrix of evaluations of the monomials $z_i z_j$, $i < j$, at a collection of points. In the first family its determinant describes the boundary of the principal component; in the second its kernel gives the transverse quadrics.

1.1 The principal component

For the first family fix

$$k \geq 3, \quad q = \binom{k}{2}, \quad h = q + k = \binom{k+1}{2}, \quad M = 2q, \quad d_0 = q(k+1) = h(k-1). \quad (1.1)$$

The symbol d_0 denotes a degree, not an ambient dimension. Put

$$b_k = \prod_{1 \leq i < j \leq q} \frac{2(k+1) + 2q - i - j}{2q - i - j}. \quad (1.2)$$

Theorem 1.1. *Work over $\mathbb{C}$, with the parameters in (1.1). The module $P(2q, h)/\mathfrak{l}(2q, h)$ is generated by its degree- d_0 part, which has dimension $2b_k$, and it has no nonzero element of smaller degree. Its generators are obtained as follows. Let G_k be the Gale-dual Veronese polynomial of Definition 3.1. As U ranges over the maximal-isotropic q -dimensional row spaces in K^{2q} , the polynomials $G_k(UV)$ span two spaces of dimension b_k , one for each component of $\text{OGr}(q, 2q)$. Their images are independent modulo $\mathfrak{l}(2q, h)$ and minimally generate $P(2q, h)/\mathfrak{l}(2q, h)$.*

The construction is finite and explicit: one takes coefficients on a dense skew-matrix chart in each orthogonal Grassmannian and extracts a basis. The first numerical cases are

k	M	h	d_0	$2b_k$
3	6	6	12	330
4	12	10	30	163324304
5	20	15	60	9627912669442441500

For $(6, 6)$, Section 4 gives all 330 equations directly as the coefficients of two determinants, without a basis-extraction step.

For the homological statements write

$$L = 2qh, \quad e = \binom{h}{2}, \quad N = L - e, \quad a_R = 2e - L. \quad (1.3)$$

All polynomial variables have degree one, and $T(s)_j = T_{j+s}$.

Theorem 1.2. *In every characteristic different from two, $C = C(2q, h)$ is normal and satisfies S_k but not S_{k+1} . Its non-Cohen–Macaulay locus is*

$$Z = \{V : V^t B V = 0, \text{ rank } V \leq q - 1\},$$

where B is a matrix of the quadratic form. This locus has codimension $k + 1$ in C , and

$$\text{depth } \mathcal{O}_{C,x} = \dim \mathcal{O}_{C,x} - 1 \quad (x \in Z).$$

Let Q_+ and Q_- be the primes of the two maximal-isotropic components, set $J = Q_+ \cap Q_-$, and put $A = S/P$ and $Z_0 = S/(Q_+ + Q_-)$. Then

$$\begin{aligned} \text{Ext}_S^e(A, S(-L)) &= (J/\mathfrak{l})(a_R), \\ \text{Ext}_S^{e+1}(A, S(-L)) &= Z_0(a_R), \end{aligned}$$

and all other Ext modules vanish. Consequently

$$\text{depth } A = N - 1, \quad \text{pd}_S A = e + 1, \quad \text{reg } A = e - 1.$$

The last nonzero module in its minimal graded free resolution is $S(-2e)$.

Thus the principal components in this family are normal but do not have rational singularities in characteristic zero and are not F -rational in odd characteristic: both properties imply Cohen–Macaulayness. The defining-ideal theorem is restricted to characteristic zero because its proof uses semisimplicity of the relevant representations. The depth and normality assertions have no such restriction.

1.2 Counterexamples at the prime threshold

The prime threshold of [2] is

$$D_{\text{prime}}(h) = \min \left\{ 2 \left\lfloor \frac{2h+1-\sqrt{8h+1}}{2} \right\rfloor + 2, 2 \left\lfloor \frac{2h+1-\sqrt{8h-7}}{2} \right\rfloor + 1 \right\}. \quad (1.4)$$

Theorem 1.3. *For $k \geq 4$, let*

$$\binom{k}{2} + 2 \leq h \leq \left\lfloor \frac{k^2}{2} \right\rfloor, \quad M = 2h - 2k + 1. \quad (1.5)$$

Then $M = D_{\text{prime}}(h)$ and $X(M, h)$ is a normal integral complete intersection. It does not have rational singularities over $\mathbb{C}$ and is not F -rational in any odd characteristic. If $h < k^2/2$, it is not log canonical over $\mathbb{C}$ and is not F -pure in any odd characteristic. The set of integers h in (1.5) has natural density $1/2$. The same density is obtained by imposing $h - \binom{k}{2} \geq 4$, in which case the coordinate rings are factorial.

This gives a negative answer to [2, Question 8.1]. The first example is $X(9, 8)$. At a generic maximal-isotropic frame it has a transverse cone over a smooth elliptic quartic. The next example, $X(15, 12)$, is already not log canonical. The generic transverse classification, including the positive cases, is given in Theorem 6.1. Those positive assertions concern the generic point of one stratum, rather than all points of the frame variety.

Theorem 1.1 answers [2, Question 8.3] for the entire family (1.1), including the stated first case. Theorem 1.2 gives negative answers to the universal Cohen–Macaulay and rationality assertions in Question 8.4 and proves normality for the same family. It does not classify principal components for arbitrary (M, h) .

2 Components, canonical modules, and a local chart

2.1 The three-component decomposition

For the parameters (1.1), the component and complete-intersection theorems of Casabella–Sammartano [2, Theorems 1 and 5.8] give

$$\mathfrak{l} = P \cap Q_+ \cap Q_-, \quad X(2q, h) = C \cup Y_+ \cup Y_-. \quad (2.1)$$

Here $Y_{\pm}$ are the closures of frames spanning maximal isotropic q -planes of the two types. The ideal $\mathfrak{l}$ is a radical complete intersection of e quadrics; all three components have dimension

$$N = qh + \frac{q(q-1)}{2} = L - e. \quad (2.2)$$

Indeed, substitution of $8h + 1 = (2k + 1)^2$ into the threshold formulas gives $D_{\text{CI}}(h) = 2q$ and $D_{\text{prime}}(h) = 2q + 1$.

We use the following additional input from Hochster–Jeffries–Pandey–Singh [6, Theorem 7.13]: the rings S/Q_+ , S/Q_- , and $Z_0 = S/(Q_+ + Q_-)$ are Cohen–Macaulay domains, and

$$Q_+ + Q_- = (V^t B V) + I_q(V), \quad \dim Z_0 = N - k - 1. \quad (2.3)$$

The parentheses in (2.3) denote all Gram entries, including the diagonal; $I_q(V)$ denotes the ideal of q -minors. Thus $\text{Spec } Z_0$ is the rank-at-most- $(q - 1)$ isotropic locus.

Canonical modules and graded local duality are used with the conventions of [1]. In particular, for an equidimensional quotient D of S of codimension e , we write $\omega_D = \text{Ext}_S^e(D, S(-L))$, without requiring D to be Cohen–Macaulay. With $R = S/\mathfrak{l}$, one has $\omega_R = R(a_R)$.

Lemma 2.1. *Let $J = Q_+ \cap Q_-$. There are graded isomorphisms*

$$\omega_{S/J} = \omega_{S/Q_+} \oplus \omega_{S/Q_-}, \quad P/\mathfrak{l} = \omega_{S/J}(-a_R). \quad (2.4)$$

In particular $P/\mathfrak{l}$ is a Cohen–Macaulay S -module of dimension N .

Proof. Apply $\text{Ext}_S^\bullet(-, S(-L))$ to

$$0 \longrightarrow S/J \longrightarrow S/Q_+ \oplus S/Q_- \longrightarrow Z_0 \longrightarrow 0. \quad (2.5)$$

The codimension of Z_0 in S is $e + k + 1$, so its Ext modules vanish in degrees e and $e + 1$. This proves the first isomorphism. The radical decomposition gives $\mathfrak{l} : J = P$. Change of rings for the complete intersection yields

$$\omega_{S/J} = \text{Hom}_R(S/J, \omega_R) = (P/\mathfrak{l})(a_R).$$

This is the canonical-module form of complete-intersection linkage [7]. Each summand in the first isomorphism is Cohen–Macaulay of dimension N , proving the last assertion. $\square$

2.2 An exact chart at an isotropic frame

The next chart will be used in both families. Write a split quadratic space as $K^q \oplus K^q \oplus K^m$ with pairing

$$\langle (u, v, z), (u', v', z') \rangle = u^t v' + v^t u' + z^t z'.$$

For $m = 0$ or 1 , the rank- q isotropic stratum is maximal isotropic.

Lemma 2.2. *Let $h = q + k$ and $M = 2q + m$. On a chart meeting the rank- q isotropic stratum, the orthogonality equations are, after an invertible change of coordinates and elimination of variables,*

$$\sum_{a=1}^m z_{ai} z_{aj} - \sum_{r=1}^q \lambda_r y_{ri} y_{rj} = 0 \quad (1 \leq i < j \leq k). \quad (2.6)$$

The coordinates $Y = (y_{ri})$ are unrestricted. In addition there are $q^2 + \binom{q}{2} + mq$ free coordinates, with a $q \times q$ determinant inverted. The isotropic stratum in this chart is $\lambda = z = 0$.

Proof. Choose q independent columns whose projection onto the first summand is invertible, and denote this top block by D . The orthogonal change $\text{diag}(D^{-1}, D^t, I_m)$ makes it the identity and retains the entries of D as independent chart coordinates. Write the columns in blocks

$$A = \begin{bmatrix} I_q \\ H \\ C \end{bmatrix}, \quad B' = \begin{bmatrix} Y \\ T \\ W \end{bmatrix}.$$

The first set of equations is $H + H^\mathbf{t} + C^\mathbf{t}C = \text{diag}(\lambda)$. It eliminates the symmetric part of H . The equations $A^\mathbf{t}BB' = 0$ for the ambient form, equivalently the pairings between columns of A and B' , give $T = -H^\mathbf{t}Y - C^\mathbf{t}W$. With $z = W - CY$, direct substitution in the Gram matrix of B' gives

$$(B')^\mathbf{t}BB' = z^\mathbf{t}z - Y^\mathbf{t}\text{diag}(\lambda)Y.$$

Its off-diagonal entries are (2.6). The changes used are invertible on the chosen chart, so this describes the local scheme, including its transverse parameters. If $\lambda = 0$, the columns of A are isotropic; $z = 0$ says that the remaining columns belong to their span. This proves the last assertion. $\square$

For a $q \times k$ matrix Y , define its hollow-quadratic evaluation matrix by

$$E(Y)_{(ij),r} = y_{ri}y_{rj}, \quad i < j. \quad (2.7)$$

It has $\binom{k}{2}$ rows and q columns. The case $m = 0$ of (2.6) is $E(Y)\lambda = 0$; when $m = 1$, it is $E(Y)\lambda = (z_iz_j)_{i < j}$.

3 The defining ideal in the triangular family

Throughout this section the field is $\mathbb{C}$ and the parameters are (1.1). Use the split form $B = \begin{bmatrix} 0 & I_q \\ I_q & 0 \end{bmatrix}$.

3.1 The Gale-dual Veronese polynomial

Let ν_2 list the $h = \binom{k+1}{2}$ quadratic monomials in k variables, in a fixed order. For an $h \times k$ matrix K_0 with rows κ_i , put

$$\Psi_k(K_0) = \det[\nu_2(\kappa_i)]_{i=1}^h. \quad (3.1)$$

The vectors in the determinant may be taken as columns; an overall sign from the ordering will be immaterial. Since $\det \text{Sym}^2(g) = \det(g)^{k+1}$,

$$\Psi_k(K_0g) = \det(g)^{k+1}\Psi_k(K_0). \quad (3.2)$$

The first fundamental theorem for SL_k expresses this polynomial in the maximal minors of K_0 ; see the invariant-theoretic discussion in [6]. Its degree in these minors is $k + 1$.

Definition 3.1. Apply the complementary-minor identification between the Plücker coordinate rings of $\text{Gr}(k, h)$ and $\text{Gr}(q, h)$ to (3.1). Evaluating the resulting brackets on a $q \times h$ matrix Z gives the *Gale-dual Veronese polynomial* $G_k(Z)$.

The definition is independent of a chosen bracket expression: the complementary-minor map respects the Plücker relations. Fixing volume forms fixes its overall sign. It gives a polynomial of degree d_0 with

$$G_k(gZ) = \det(g)^{k+1}G_k(Z). \quad (3.3)$$

It vanishes on matrices of rank less than q .

Lemma 3.2. *On the chart $Z = A[I_q \mid Y]$, with A invertible,*

$$G_k(Z) = \det(A)^{k+1} \det E(Y), \quad (3.4)$$

after fixing the sign in Definition 3.1. In particular G_k is nonzero.

Proof. A basis matrix for the kernel of $[I_q \mid Y]$ is $[-Y^t \mid I_k]^t = [-Y; I_k]$. In its Veronese determinant the last k rows evaluate the squares at the coordinate points. Eliminating the square columns leaves the matrix of the mixed monomials on the first q rows, namely $E(Y)$ up to transpose and sign. Equation (3.3) gives the factor $\det(A)^{k+1}$. If the rows of Y are the vectors $e_i + e_j$, one for each pair $i < j$, then $E(Y)$ is a permutation matrix. This proves nonvanishing. $\square$

Proposition 3.3. *For every rank- q matrix U satisfying $UB^{-1}U^t = 0$, the polynomial $G_k(UV)$ belongs to P .*

Proof. Complete U to hyperbolic coordinates. In these coordinates write $V = [Z^t \ W^t]^t$ with $Z = UV$. On the dense open subset defining the principal component,

$$D = V^t B V = Z^t W + W^t Z$$

is invertible and diagonal. If $\text{rank } Z = q$, let K_0 be a basis matrix for $\ker Z$. Then $K_0^t D K_0 = 0$ gives a linear relation among the h vectors $\nu_2(\kappa_i)$, with nonzero coefficients D_{ii} . Hence $\Psi_k(K_0) = 0$, and complementary minors give $G_k(Z) = 0$. If $\text{rank } Z < q$, the same vanishing follows from the bracket definition. Taking closures proves the assertion. $\square$

3.2 Canonical generators of the isotropic components

Let $X_\pm = \text{OGr}(q, 2q)^\pm$ and let $\mathcal{U}$ be the tautological rank- q bundle on either component. There is a proper birational map

$$\pi : \text{Tot}(\mathcal{U} \otimes (\mathbb{C}^h)^*) \longrightarrow Y_\pm. \quad (3.5)$$

It is an isomorphism over the rank- q locus. The rank-at-most- $(q-1)$ locus in the total space has codimension $h - q + 1 = k + 1$; its image has the same codimension by (2.3). Since $Y_\pm$ is Cohen–Macaulay and smooth on the rank- q locus, it is normal. The complement of this common open set has codimension at least two on both sides of (3.5). Extension of functions and of sections of reflexive canonical sheaves therefore gives

$$\mathbb{C}[Y_\pm] = \Gamma(\text{Tot}(\mathcal{U} \otimes (\mathbb{C}^h)^*), \mathcal{O}), \quad \omega_{\mathbb{C}[Y_\pm]} = \Gamma(\text{Tot}(\mathcal{U} \otimes (\mathbb{C}^h)^*), \omega). \quad (3.6)$$

No rational-singularity assertion is needed for this step.

The tangent bundle of $X_\pm$ is $\bigwedge^2 \mathcal{U}^*$, so

$$\omega_{X_\pm} = (\det \mathcal{U})^{q-1}, \quad \omega_{\text{Tot}(\mathcal{U} \otimes (\mathbb{C}^h)^*)} = \pi^*(\det \mathcal{U}^*)^{k+1}. \quad (3.7)$$

For the grading there is also a shift by the fiber dimension qh : the differential of each fiber coordinate has degree one. Thus, with $A_\pm = S/Q_\pm$ and $a = k + 1$,

$$(A_\pm)_m = H^0(X_\pm, \text{Sym}^m(\mathcal{U}^* \otimes \mathbb{C}^h)), \quad (3.8)$$

$$(\omega_{A_\pm})_{m+qh} = H^0(X_\pm, \text{Sym}^m(\mathcal{U}^* \otimes \mathbb{C}^h) \otimes (\det \mathcal{U}^*)^a), \quad m \geq 0. \quad (3.9)$$

There are no canonical sections in degrees below qh .

For a partition λ of length at most q , the Cauchy decomposition and the dominant-weight Borel–Weil formula give

$$(A_\pm)_m = \bigoplus_{\lambda \vdash m} \mathbf{S}_\lambda \mathbb{C}^h \otimes V_{\lambda, \pm}, \quad (3.10)$$

$$(\omega_{A_\pm})_{m+qh} = \bigoplus_{\lambda \vdash m} \mathbf{S}_\lambda \mathbb{C}^h \otimes V_{\lambda+(a^q), \pm}. \quad (3.11)$$

Here the type- D_q weights are $(\lambda_1, \dots, \lambda_q)$ and $(\lambda_1, \dots, \lambda_{q-1}, -\lambda_q)$ on the two components, with the same sign convention after the shift. The last-coordinate convention can interchange the labels $+$ and $-$, which has no effect on the arguments. The homogeneous-bundle formulas and the spinor line-bundle convention are reviewed in [8, Sections 2.2, 3.3, and 4.1]. All the displayed weights are dominant. The formulas apply to each Cauchy summand and do not impose an upper bound on the number h of frame columns.

Lemma 3.4. *The module $\omega_{A_{\pm}}$ is generated in degree qh , by b_k elements.*

Proof. Multiply sections in (3.10) by $H^0(X_{\pm}, (\det \mathcal{U}^*)^a)$. On the orthogonal-group factor for a fixed λ this is an equivariant map to the irreducible module $V_{\lambda+(a^q), \pm}$. It is nonzero: multiplication of a nonzero bundle section by a nonzero scalar section on the integral variety $X_{\pm}$ is nonzero. Hence it is surjective. The GL_h factors separate the different partitions, proving generation in the initial degree. The number of generators is $\dim H^0(X_{\pm}, (\det \mathcal{U}^*)^a)$. In the Weyl dimension formula for the weight $(a, \dots, a)$ the factors from roots $e_i - e_j$ are one, whereas the factors from $e_i + e_j$ are $(2a + 2q - i - j)/(2q - i - j)$. This is (1.2). $\square$

Lemma 2.1 now shows that $P/\mathfrak{l}$ is generated in degree

$$qh + a_R = h(k - 1) = d_0$$

and that its initial component has dimension $2b_k$.

3.3 The explicit equations span the generators

For fixed V , the function $U \mapsto G_k(UV)$ transforms by $\det(g)^{k+1}$ under a change of row basis $U \mapsto gU$. It therefore defines a section of $(\det \mathcal{U}^*)^{k+1}$ on each $X_{\pm}$. Its coefficient polynomials in V induce equivariant maps

$$H^0(X_{\pm}, (\det \mathcal{U}^*)^{k+1})^* \longrightarrow (P/\mathfrak{l})_{d_0}. \quad (3.12)$$

Each source is irreducible of dimension b_k . The map is nonzero: take $U = [I_q \mid 0]$ and $V = [Z^t \ 0]^t$, where $G_k(Z) = 1$ as in Lemma 3.2. All Gram entries of V vanish, but $G_k(UV) = 1$. The orthogonal symmetry interchanging the two Grassmannian components supplies the same witness for the other map. Thus both maps are injective.

Their images are independent modulo $\mathfrak{l}$. Fix a maximal isotropic column space S_0 . Exactly one component of maximal isotropic kernels contains planes transverse to S_0 ; every plane in the other component meets S_0 nontrivially. On the latter component $\mathrm{rank}(U|_{S_0}) < q$, so $G_k(UV) = 0$ for every frame in S_0 . As S_0 ranges through its entire Grassmannian component, this says that one coefficient module vanishes on the corresponding $Y_{\pm}$. The other has a nonzero restriction there, by the preceding witness. Restriction to the whole $Y_{\pm}$ is SO_{2q} -equivariant, so irreducibility makes that restriction injective. Interchanging the two types proves independence. The two modules consequently fill the $2b_k$ -dimensional initial component of $P/\mathfrak{l}$, completing the proof of Theorem 1.1.

Remark 3.5. The use of the whole isotropic frame component in this argument is essential: restriction to frames in one fixed plane is not equivariant for the full orthogonal group. The generation statement follows from canonical modules, rather than from a finite calculation of equations that happen to vanish on orthogonal frames.

4 The six-by-six ideal

For $a = (a_1, a_2, a_3)$, let

$$A(a) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}, \quad \nu(z) = (z_1^2, z_2^2, z_3^2, z_1 z_2, z_1 z_3, z_2 z_3)^t.$$

Use the split form $B = \begin{bmatrix} 0 & I_3 \\ I_3 & 0 \end{bmatrix}$. For a 6×6 matrix $V = [v_1 \cdots v_6]$, define

$$F_+(a, V) = \det[\nu([I_3 \mid A(a)]v_i)]_{i=1}^6, \quad (4.1)$$

$$F_-(a, V) = \det[\nu([A(a) \mid I_3]v_i)]_{i=1}^6. \quad (4.2)$$

Theorem 4.1. *Each polynomial $F_\pm$ has degree at most eight in a . Write*

$$F_\pm(a, V) = \sum_{|\alpha| \leq 8} a^\alpha f_{\pm, \alpha}(V).$$

Then $P(6, 6)$ is minimally generated by the fifteen off-diagonal Gram quadrics and the 330 polynomials $f_{\pm, \alpha}$, each of degree twelve.

Proof. First, the polynomials vanish on the principal component. On an orthogonal basis let $D = V^t B V$; then $VD^{-1}V^t = B^{-1}$. For either isotropic matrix U in (4.1)–(4.2), this gives

$$\sum_i D_{ii}^{-1} (Uv_i)(Uv_i)^t = 0.$$

The six Veronese columns are therefore dependent.

To check the parameter degree, homogenize the first matrix to $[a_0 I_3 \mid A(a)]$. The determinant is homogeneous of degree twelve in (a_0, a) . Over $\mathbb{C}(a_1, a_2, a_3)$, choose an invertible row change that makes the third row of $A(a)$ zero. The third output coordinate is then divisible by a_0 , so the rows corresponding to $z_3^2, z_1 z_3, z_2 z_3$ contribute a factor a_0^4 . The determinant of the row change on quadratic coordinates is the fourth power of its determinant and does not involve a_0 . Divisibility by a_0^4 over the fraction field implies divisibility in the polynomial ring. Dehomogenizing proves the degree bound. The other family is obtained by interchanging the two coordinate blocks. There are $\binom{11}{3} = 165$ possible coefficients in each family.

For independence, set

$$Z = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}, \quad V_+(t) = \begin{bmatrix} Z \\ A(t)Z \end{bmatrix}, \quad V_-(t) = \begin{bmatrix} A(t)Z \\ Z \end{bmatrix}.$$

These are isotropic frames, and the Veronese determinant of Z is one. The identity $A(a)A(t) = ta^t - (a \cdot t)I_3$ gives $\det(I_3 + A(a)A(t)) = (1 - a \cdot t)^2$. Since $\det \text{Sym}^2(g) = \det(g)^4$, one obtains

$$\begin{aligned} F_+(a, V_+(t)) &= (1 - a \cdot t)^8, & F_-(a, V_+(t)) &= 0, \\ F_-(a, V_-(t)) &= (1 - a \cdot t)^8, & F_+(a, V_-(t)) &= 0. \end{aligned}$$

The zero entries follow from $\text{rank}(A(a) + A(t)) \leq 2$. For $|\alpha| \leq 8$, the coefficient of a^α in the nonzero expression is

$$(-1)^{|\alpha|} \frac{8!}{(8 - |\alpha|)! \alpha_1! \alpha_2! \alpha_3!} t^\alpha.$$

These are nonzero multiples of distinct monomials over $\mathbb{C}$. Evaluation on the two frame families proves that all 330 classes are independent modulo $\mathfrak{l}(6, 6)$. Theorem 1.1 gives $d_0 = 12$ and $2b_3 = 330$, so they form the entire initial generating space. This proves the assertion and minimality. $\square$

4.1 Hilbert series and the real vanishing ideal

The canonical-module calculation is particularly concrete here. One component of $\text{OGr}(3, 6)$ is $\mathbb{P}^3$: identify the quadratic space with $\wedge^2 \mathbb{C}^4$ and parameterize its maximal isotropic planes by lines in $\mathbb{C}^4$. If $\mathcal{Q}$ is the quotient bundle, then $\mathcal{U} = \mathcal{O}(-1) \otimes \mathcal{Q}$ and $\det \mathcal{U} = \mathcal{O}(-2)$. The total-space canonical bundle in (3.7) is the pullback of $\mathcal{O}(8)$, with fiber-degree shift eighteen. This also gives $h^0(\mathbb{P}^3, \mathcal{O}(8)) = 165$ canonical generators on each isotropic component.

Corollary 4.2. *Over $\mathbb{C}$, the Hilbert series of an isotropic component ring $A_{\pm}$ and of the principal component ring $A = S/P(6, 6)$ are*

$$H_{A_{\pm}}(t) = \frac{1 + 15t + 99t^2 + 165t^3}{(1 - t)^{21}},$$

$$H_A(t) = \frac{(1 + t)^{15} - 2t^{12}(165 + 99t + 15t^2 + t^3)}{(1 - t)^{21}}.$$

In particular, $\deg C(6, 6) = 32208$.

Proof. The initial canonical degree is eighteen, so the Cohen–Macaulay ring $A_{\pm}$ has a -invariant -18 and regularity $21 - 18 = 3$. Its Hilbert numerator therefore has degree three. For completeness, on $\mathbb{P}^3$ the summands in degree m have dimensions

$$\dim \mathbf{S}_{\lambda} \mathbb{C}^6 \cdot \dim \mathbf{S}_{(m, \lambda_1, \lambda_2, \lambda_3)} \mathbb{C}^4, \quad \lambda \vdash m, \quad \ell(\lambda) \leq 3.$$

The Weyl dimension formula in degrees zero through three gives 1, 36, 645, 7480, respectively. Multiplication by $(1 - t)^{21}$ gives the first numerator. Canonical duality reverses that numerator, with the canonical degree shift eighteen. Lemma 2.1 then places the two reversed numerators in degree twelve inside R , whose Hilbert series is $(1 + t)^{15}/(1 - t)^{21}$. Subtracting gives the second formula. Its numerator at one is $2^{15} - 2(165 + 99 + 15 + 1) = 32208$. $\square$

Corollary 4.3. *For the positive definite real quadratic form, the real vanishing ideal of pairwise orthogonal h -frames in the family (1.1) is the real form of $P(2q, h)$. In particular, Theorem 4.1, after a complex linear change of the ambient quadratic coordinates and real descent, determines all polynomial equations of orthogonal bases of $\mathbb{R}^6$ and their closure.*

Proof. The nonzero columns of a real orthogonal frame are independent. Since $M \geq h$, any zero columns can be replaced by arbitrarily small nonzero vectors in independent orthogonal directions. Thus every real frame is a limit of the open locus defining the principal component. Conversely, that open real locus is Zariski dense in the complex principal component: a standard real orthogonal frame is a smooth point, and its real smooth neighborhood has real dimension equal to the complex dimension of the component. The analytic identity theorem gives the required density. The complex ideal is invariant under conjugation and therefore descends to the stated real ideal. $\square$

5 Depth, deficiency, and normality

We return to an arbitrary algebraically closed field of characteristic different from two, retaining (1.1) and (1.3). The arguments in this section do not use the characteristic-zero representation decomposition.

5.1 The deficiency module

Proposition 5.1. *For $A = S/P$, the only nonzero Ext modules into $S(-L)$ are*

$$\mathrm{Ext}_S^e(A, S(-L)) = (J/I)(a_R), \quad \mathrm{Ext}_S^{e+1}(A, S(-L)) = Z_0(a_R). \quad (5.1)$$

Proof. Set $T = P/I$. Lemma 2.1 says that T and $R = S/I$ are Cohen–Macaulay modules of dimension N . Apply $\mathrm{Ext}_S^\bullet(-, S(-L))$ to

$$0 \longrightarrow T \longrightarrow R \longrightarrow A \longrightarrow 0.$$

Canonical duality on the two Cohen–Macaulay component rings gives

$$0 \longrightarrow \omega_A \longrightarrow R(a_R) \longrightarrow (A_+ \oplus A_-)(a_R) \longrightarrow \mathrm{Ext}_S^{e+1}(A, S(-L)) \longrightarrow 0, \quad (5.2)$$

and all other Ext modules vanish.

After removing the common shift, each component of the middle map is a graded R -linear map $R \rightarrow A_\pm$. Its value on one is a scalar. The scalar is nonzero, since at the generic point of $Y_\pm$ the inclusion $T \rightarrow R$ is an isomorphism. Normalize the two canonical identifications to make both scalars one. The middle map is then the pair of quotient maps. Its kernel is J/I and its cokernel is Z_0 , by (2.5). This proves (5.1). $\square$

Corollary 5.2. *The locus $Z = \mathrm{Spec} Z_0$ is contained in C . The ring A is Cohen–Macaulay away from Z , and at every $x \in Z$ it has local depth $\dim \mathcal{O}_{C,x} - 1$. It satisfies S_k but not S_{k+1} .*

Proof. The support of an Ext module is contained in the support of its first argument. The second module in (5.1) therefore proves $Z \subset C$. Localize in the regular ring S at a prime lying on C . If it is not on Z , the projective dimension of A there is e ; if it is on Z , the nonzero last Ext module makes that projective dimension $e + 1$. The codimension of the principal prime remains e under this localization. Auslander–Buchsbaum gives the two asserted depths.

The locus Z has codimension $k + 1$ by (2.3). Thus at points of Z the local dimension is at least $k + 1$, so the depth is at least k . Elsewhere the ring is Cohen–Macaulay. This proves S_k . At the generic point of Z , local dimension is $k + 1$ and depth is k , proving failure of S_{k+1} . $\square$

Proposition 5.3. *The global invariants and last free module are*

$$\mathrm{depth} A = N - 1, \quad \mathrm{pd}_S A = e + 1, \quad \mathrm{reg} A = e - 1, \quad F_{e+1} = S(-2e).$$

Moreover, $\mathrm{depth} \omega_A = N - k + 1$.

Proof. The first two assertions follow from Proposition 5.1. The module $Z_0(a_R)$ is cyclic, with its generator in degree $-a_R$. Dualizing a minimal graded free resolution preserves minimality of the differentials. Thus the last free module has rank one. If its shift is j , its dual into $S(-L)$ has generator degree $L - j$. Equation (5.1) gives $L - j = -a_R$, hence $j = 2e$.

The initial degree of J/I is two. Indeed, the diagonal norm quadrics belong to J and are independent of the off-diagonal quadrics by column multidegrees. There are no linear equations on the isotropic union: in split coordinates every coordinate matrix unit is an isotropic frame. Graded local duality therefore gives

$$\mathrm{end} H_{\mathfrak{m}}^{N-1}(A) = a_R, \quad \mathrm{end} H_{\mathfrak{m}}^N(A) = a_R - 2,$$

and all other local cohomology modules vanish. Since $a_R + N = e$, their degree formula for regularity gives $\mathrm{reg} A = e - 1$.

From (2.5) one has $\mathrm{depth}(S/J) = N - k$. The sequence $0 \rightarrow J/I \rightarrow R \rightarrow S/J \rightarrow 0$ then gives $\mathrm{depth}(J/I) = N - k + 1$. Apply the first isomorphism in (5.1) to obtain the last assertion. $\square$

For $(M, h) = (6, 6)$, these formulas give dimension 21, depth 20, regularity 14, projective dimension 16, and last free module $S(-30)$. Its canonical module has depth 19.

5.2 The boundary determinant

Corollary 5.2 gives S_2 . To prove normality it remains to prove regularity in codimension one.

Lemma 5.4. *Let $q = \binom{k}{2}$, $k \geq 3$, and set $D_k(Y) = \det E(Y)$ for a $q \times k$ matrix Y . Then D_k is irreducible and its generic zero has $\text{rank } E(Y) = q - 1$. At a general point of $D_k = 0$, the relation $E(Y)\lambda = 0$ has all $\lambda_r \neq 0$, and*

$$Y^t \text{diag}(\lambda)Y = \text{diag}(\mu_1, \dots, \mu_k) \quad \text{has all } \mu_j \neq 0.$$

Proof. Let H be the q -dimensional space spanned by the hollow quadrics $x_i x_j$, $i < j$. Define the incidence scheme

$$\mathcal{Z} = \{([Q], y_1, \dots, y_q) \in \mathbb{P}(H) \times (\mathbb{A}^k)^q : Q(y_r) = 0 \text{ for all } r\}.$$

It is the q -fold fiber product of the universal affine quadric over $\mathbb{P}(H)$. To check flatness, put $W_m = K[x_1, \dots, x_k]_m$, with $W_m = 0$ for $m < 0$. Its degree- m coordinate module is the cokernel of multiplication by the universal quadric,

$$\mathcal{O}_{\mathbb{P}(H)}(-1) \otimes_K W_{m-2} \longrightarrow \mathcal{O}_{\mathbb{P}(H)} \otimes_K W_m.$$

Every fiber map is injective, since its defining quadratic is nonzero. The map has constant rank, so its cokernel is locally free. Taking the direct sum over m proves flatness of the universal affine quadric, and hence of $\mathcal{Z}$ over $\mathbb{P}(H)$.

The hollow quadric $x_1 x_2 + x_1 x_3 + x_2 x_3$ has rank three in every characteristic different from two. Thus a general hollow quadric has rank at least three and is geometrically integral, as is its q -fold product. Flatness over the integral base makes the generic fiber dense and makes each affine coordinate ring inject into its generic-fiber algebra, which is a domain. It follows that $\mathcal{Z}$ is integral.

Projection to Y is proper because the other factor is projective. A point is in its image precisely when there is a nonzero hollow quadric vanishing at all its rows, or equivalently when $\ker E(Y)^t \neq 0$. The image is therefore exactly $D_k = 0$, which is irreducible as a closed set. The choice $y_r = e_i + e_j$, one row for each pair, makes $E(Y)$ the identity. Thus D_k is nonzero and is a scalar times a power of an irreducible polynomial.

We show that this power is one. Fix a general integral quadric Q , with coefficient vector $a \in H$. Evaluations of H on Q span $a^\perp$: the only degree-two polynomial in H vanishing on Q is a scalar multiple of Q . Choose q smooth points of Q such that every $q - 1$ of their evaluation vectors are independent. Such tuples form a nonempty open subset of the irreducible product $V(Q)^q$, since each of the finitely many independence conditions holds on a nonempty open subset. At this tuple, E has rank $q - 1$. Its left kernel is spanned by a , and its right kernel is spanned by a vector λ with every $\lambda_r \neq 0$. Consequently

$$\text{adj}(E) = \gamma \lambda a^t \quad \text{for some } \gamma \neq 0.$$

If only the row y_r moves, in the direction v , differentiation gives

$$dD_k(Y)(0, \dots, v, \dots, 0) = \gamma \lambda_r dQ_{y_r}(v).$$

Choose v transverse to Q at y_r . The right side is nonzero, so D_k is reduced at this point and cannot be a proper power. This proves irreducibility as a polynomial and also the generic rank and full-support assertions.

For each j , the restrictions of $H + \langle x_j^2 \rangle$ to Q have dimension q : their only relation is Q , and x_j^2 is not hollow. Evaluation at q general points is an isomorphism on this restriction space. The relation λ among the hollow evaluations therefore does not annihilate x_j^2 , so

$$\mu_j = \sum_r \lambda_r y_{rj}^2 \neq 0.$$

These finitely many evaluation conditions can be imposed together with the preceding independence conditions. They give a nonempty open subset of $D_k = 0$ on which all the stated conclusions hold. $\square$

Proposition 5.5. *The principal component is smooth at the generic point of every divisor it has in the rank- q isotropic stratum.*

Proof. Use Lemma 2.2 with $m = 0$. The equations are $E(Y)\lambda = 0$ and the isotropic component is $\lambda = 0$. At a general point of $D_k = 0$, invert a $(q-1)$ -minor and eliminate $q-1$ of the λ coordinates. The remaining equation is

$$D_k(Y)u = 0$$

up to a unit. The branch $u = 0$ is isotropic. On the other branch, Lemma 5.4 gives points with $u \neq 0$ for which all first q norms λ_r and all remaining norms $-\mu_j$ are nonzero. Thus the branch $D_k = 0$ is the principal component. Where $dD_k \neq 0$, it is smooth also at $u = 0$, its intersection with the isotropic branch.

For completeness, this accounts for every divisor in the indicated stratum. If D_k is a unit, the chart has only the isotropic component, so no point there belongs to C . On the isotropic stratum itself, where $\lambda = 0$, the possible intersection with C is therefore supported on $D_k = 0$. The preceding calculation places the generic point of this irreducible hypersurface in C ; closedness places its whole closure there. The locus where the rank, nonvanishing, or smoothness conditions used above fail is a proper closed subset of that hypersurface. Since the hypersurface has dimension $N-1$ after adjoining the free chart coordinates, the exceptional subset has dimension at most $N-2$. The column and Witt charts cover the rank- q isotropic stratum, proving the assertion. $\square$

5.3 The remaining codimension-one strata

For a frame let t be the number of columns of nonzero norm and r the rank of its isotropic columns. For a fixed choice of the t indices, the stratum has dimension

$$D(t, r) = Mt - \frac{t(t-1)}{2} + r(M + h - 2t - r) - \frac{r(r+1)}{2}, \quad t + r \leq h, \quad t + 2r \leq M. \quad (5.3)$$

The first term chooses the nonisotropic orthogonal columns. In their nondegenerate orthogonal complement, the orthogonal Grassmannian of isotropic r -planes has dimension $r(M - t - r) - r(r+1)/2$. A spanning $(h-t)$ -tuple in the chosen plane contributes $r(h-t)$ parameters. These choices give (5.3); every component of the stratum has this dimension.

Suppose $k \geq 4$, put $x = q - r$, and write $c_x(t) = N - D(t, q - x)$. At our parameters,

$$c_x(t) = \frac{2kx + t^2 - 4tx - t + 3x^2 - x}{2}. \quad (5.4)$$

The constraints become $0 \leq t \leq \min(2x, k+x)$, and

$$c_x(t+1) - c_x(t) = t - 2x.$$

Thus the codimension decreases through the allowed successive values. If $1 \leq x \leq k$, set $t = 2x - s$. Then

$$c_x(t) = \frac{x(2k - x - 3)}{2} + \frac{s(s+1)}{2}.$$

The first term is concave in x ; its endpoint values are $k-2$ and $k(k-3)/2$, both at least two. None of these strata can contribute a divisor.

If $x \geq k$, set $t = k + x - s$ and $r = q - x$. Now

$$c_x(t) = r + \frac{s(2x - 2k + s + 1)}{2}.$$

For $r \geq 2$ this is at least two. For $r = 1$, the value $s = 0$ gives the codimension-one stratum $t = h - 1$, whose frames have rank $t + r = h$; if $s \geq 1$, the codimension is at least $1 + q - k \geq 3$. For $r = 0$, the value $s = 0$ is the full-rank open stratum, whereas $s \geq 1$ gives codimension at least $q - k + 1 \geq 3$. Finally, $x = 0$ forces $t = 0$ and $r = q$, the maximal-isotropic stratum treated in Proposition 5.5.

A full-column-rank frame is smooth on $\mathbf{X}$: a relation among the Jacobian rows is a hollow symmetric matrix K_0 with $VK_0 = 0$, which forces $K_0 = 0$. This treats all remaining divisors for $k \geq 4$.

For $k = 3$, one has $q = 3$, $h = 6$, and $N = 21$. Substitution in (5.3) gives the complete list of strata with $N - D(t, r) \leq 1$:

t	r	$N - D(t, r)$	rank $V = t + r$
0	3	0	3
2	2	1	4
4	1	1	5
5	0	1	5
6	0	0	6

The first and last rows have already been treated. In the $(4, 1)$ and $(5, 0)$ cases, $\ker V$ is one-dimensional. Every symmetric Jacobian relation has image in this kernel and therefore has the form buu^t for a generator u . Some coordinate u_i is nonzero, and the zero diagonal gives $bu_i^2 = 0$, hence $b = 0$.

For $(t, r) = (2, 2)$, split off the two nonisotropic columns. Write the remaining four columns in a basis of their isotropic two-plane. For an explicit rank-two coordinate matrix use

$$[I_2 \mid Y], \quad Y = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The first two coordinates of any relation among all six columns are zero, because the nonisotropic span meets its orthogonal complement trivially. The residual kernel has basis matrix $W = [-Y; I_2]$, with rows

$$(-1, -1), \quad (-1, 1), \quad (1, 0), \quad (0, 1).$$

A symmetric relation on the residual four columns has the form WTW^t , where $T = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$. Its last two diagonal entries force $a = c = 0$, and its first diagonal entry then forces $2b = 0$. Since the characteristic is different from two, $T = 0$, so there is no nonzero Jacobian relation at this frame.

This coordinate matrix can be placed in every isotropic two-plane. For each such plane, the spanning four-tuples form an irreducible open subset of $\text{Mat}_{2,4}$, and full Jacobian rank is a nonempty open condition there. The argument applies to both isotropic-plane types in each fixed orthogonal complement. The bad locus has positive codimension in every family of spanning four-tuples, so it has dimension at most $N - 2$ in the stratum. Thus it contributes no divisor of C .

We have proved R_1 in all cases. Together with S_2 from Corollary 5.2, Serre's criterion proves normality. Propositions 5.1 and 5.3 now complete the proof of Theorem 1.2.

6 Generic transverse singularities in odd ambient dimension

We now use a different parameter range. Let

$$k \geq 2, \quad 2 \leq c \leq k, \quad h = \binom{k}{2} + c, \quad q = h - k, \quad M = 2q + 1. \quad (6.1)$$

Formula (1.4) gives $M = D_{\text{prime}}(h)$. Let S_{iso} be the stratum of frames spanning a maximal isotropic q -plane. It is irreducible, and its dimension and codimension in $\mathbf{X}(M, h)$ are

$$\dim S_{\text{iso}} = qh + \frac{q(q+1)}{2}, \quad \text{codim}_{\mathbf{X}(M, h)} S_{\text{iso}} = c. \quad (6.2)$$

The latter follows by subtracting from $Mh - \binom{h}{2}$.

Theorem 6.1. *Let L be the function field of S_{iso} , and set*

$$\delta = k - c, \quad a = 2\delta - k = k - 2c = k^2 - 2h.$$

At the generic point of S_{iso} , the local ring of $\mathbf{X}(M, h)$ is

$$(L[z_1, \dots, z_k]/(Q_1, \dots, Q_\delta))_{(z)}, \quad (6.3)$$

where the Q_i are hollow quadrics and their projective intersection is geometrically smooth. If $\delta = 0$, this local ring is regular. For $\delta > 0$ the classification is

Condition	Characteristic zero	Odd characteristic
$a < 0$	rational and canonical	strongly F -regular
$a = 0$	log canonical, not rational	F -pure, not F -rational
$a > 0$	not log canonical	not F -pure

Every assertion in this table concerns the generic transverse local ring.

6.1 Elimination and dominance

Apply Lemma 2.2 with $m = 1$ and write the last row of z as $(z_1, \dots, z_k)$. The residual equations are

$$z_i z_j - \sum_{r=1}^q \lambda_r y_{ri} y_{rj} = 0, \quad i < j. \quad (6.4)$$

Put $e' = \binom{k}{2}$, so $e' - q = \delta$. The matrix $E(Y)$ has generic rank q : choose its q rows of Y among distinct vectors $e_i + e_j$, so its columns are distinct coordinate vectors. At the generic point of the

stratum, q equations eliminate the λ_r . The remaining quadrics form the kernel of $E(Y)^\dagger$ in the e' -dimensional space

$$H = \langle z_i z_j : i < j \rangle.$$

All chart parameters other than λ, z become elements of L . Consequently this is the exact local presentation (6.3), not only an associated graded ring.

Lemma 6.2. *If $\delta > 0$, the rational map*

$$\mathrm{Mat}_{q,k} \dashrightarrow \mathrm{Gr}(\delta, H), \quad Y \longmapsto \ker E(Y)^\dagger$$

is dominant. A general δ -dimensional subspace of H cuts out a geometrically smooth projective complete intersection of dimension $c - 1$.

Proof. We first show that the smooth locus in the indicated Grassmannian is nonempty. Choose δ diagonal quadrics $\sum_i b_{ji} z_i^2$ such that every maximal minor of the $\delta \times k$ matrix (b_{ji}) is nonzero. Vandermonde columns over the infinite field supply such a matrix. Any common projective zero has at least $\delta + 1$ nonzero coordinates; otherwise a full-column-rank submatrix of (b_{ji}) would annihilate the nonzero vector of their squares. The Jacobian therefore has rank δ at every common zero, because two is invertible. The intersection is a nonempty smooth complete intersection of dimension $k - 1 - \delta = c - 1 \geq 1$.

The Koszul resolution shows that it is connected and has no linear equations. Smoothness and connectedness make it integral. It contains k linearly independent projective points. Send these points to the coordinate points by a projective change of coordinates. The resulting quadrics are hollow and still define a smooth complete intersection. Smoothness of such an intersection is an open condition on the Grassmannian, proving the required nonempty open subset.

Now let $K_0 \subset H$ lie in that open subset, and let T be its projective zero set. Its degree-two ideal is exactly K_0 , since it is a positive-dimensional complete intersection of those quadrics. It follows that the hollow evaluation vectors $(z_i z_j)_{i < j}$ at points of T span $K_0^\perp$, a vector space of dimension q . Choose q points with independent evaluations and use representatives as the rows of Y . Then $E(Y)$ has rank q and $\ker E(Y)^\dagger = K_0$. The image contains the whole nonempty open subset, proving dominance. These arguments also apply after algebraically closed extensions, giving geometric smoothness. $\square$

The quadrics in (6.3) are consequently a regular sequence, and their graded ring is a normal complete intersection of dimension $c \geq 2$, singular only possibly at the vertex. Its a -invariant is $2\delta - k = a$.

6.2 Characteristic zero

Suppose $\delta > 0$, and let $T \subset \mathbb{P}_L^{k-1}$ be the smooth intersection from Lemma 6.2. Its cone is resolved by blowing up the vertex; the resolution is $\mathrm{Tot}(\mathcal{O}_T(-1))$. Adjunction gives $\omega_T = \mathcal{O}_T(a)$. If E is the exceptional divisor, then $\mathcal{O}_E(E) = \mathcal{O}_T(-1)$, and the discrepancy is

$$-a - 1. \tag{6.5}$$

For example, this follows by restricting $(K_{\mathrm{Tot}(\mathcal{O}_T(-1))} + E)|_E = K_T$ and using that the cone is Gorenstein. The resolution has one smooth exceptional divisor. Thus the cone is log canonical exactly when $a \leq 0$, and canonical exactly when $a < 0$.

Rationality can be read off directly. Since T is an arithmetically Cohen–Macaulay complete intersection, its intermediate cohomology $H^i(T, \mathcal{O}_T(m))$ vanishes for $m \geq 0$ and $0 < i < \dim T$. By Serre duality its top cohomology is

$$H^{c-1}(T, \mathcal{O}_T(m)) = H^0(T, \mathcal{O}_T(a-m))^*.$$

All these groups vanish for $m \geq 0$ exactly when $a < 0$. If $a \geq 0$, the group at $m = 0$ is nonzero. The cohomology of the total-space resolution is the direct sum of these groups over $m \geq 0$, which computes the higher direct images because the cone is affine. This proves rationality precisely for $a < 0$ and establishes the characteristic-zero column of the table.

6.3 Odd characteristic

We recall the complete-intersection Frobenius criteria in the form needed here. Let $R = L[z]_{(z)}/(Q_1, \dots, Q_\delta)$, where L is F -finite, and put $F = Q_1 \cdots Q_\delta$. Fedder’s criterion [3] says that R is F -pure if and only if

$$F^{p-1} \notin (z_1^p, \dots, z_k^p). \quad (6.6)$$

If u is a nonzero element for which R_u is strongly F -regular, Glassbrenner’s criterion [4, Theorem 4.1] gives strong F -regularity from

$$uF^{p^e-1} \notin (z_1^{p^e}, \dots, z_k^{p^e}) \quad \text{for some } e \geq 1. \quad (6.7)$$

Conversely, in a strongly F -regular ring this nonmembership occurs for every nonzero u and a suitable e . These forms follow from the usual complete-intersection colon-ideal formula. Our residue field L is finitely generated over the perfect field K , hence is F -finite. The cone is regular away from its vertex, so localization at a nonzero linear coordinate is regular.

Suppose $2\delta < k$. Work first with arbitrary independent hollow quadrics as parameters. At the special system

$$Q_i = z_{2i}z_{2i+1}, \quad 1 \leq i \leq \delta,$$

the coefficient of $z_1 \prod_{j=2}^{2\delta+1} z_j^{p-1}$ in $z_1 F^{p-1}$ is one. It is a polynomial in the coefficients of the Q_i , so it is generically nonzero. Every exponent is below p . The nonvanishing locus meets the open locus of smooth projective complete intersections from Lemma 6.2. On this intersection, apply (6.7) with $u = z_1$. Dominance then proves strong F -regularity of the generic transverse ring. The monomial system is only a witness that a coefficient polynomial is nonzero; no smoothness assertion about that special system is used.

If $2\delta = k$, use the special system $Q_i = z_{2i-1}z_{2i}$. The coefficient of $\prod_j z_j^{p-1}$ in F^{p-1} is one at that system and hence nonzero generically. This proves F -purity by (6.6). For every p -power Q and nonzero linear coordinate u ,

$$\deg(uF^{Q-1}) = 1 + k(Q-1) > k(Q-1).$$

Every monomial therefore lies in $(z_1^Q, \dots, z_k^Q)$, contradicting the necessary direction of (6.7). The ring is not strongly F -regular. An F -rational F -finite Gorenstein local ring is strongly F -regular [5, Corollary 4.7 and Theorem 5.5]; thus it is not F -rational.

Finally, if $2\delta > k$, the same degree comparison already places F^{p-1} in $(z_1^p, \dots, z_k^p)$. It is not F -pure, and again cannot be F -rational. This proves the remaining column of Theorem 6.1. Changes of basis among the quadrics do not affect these criteria, so the nonempty open conditions used above descend from the space of ordered systems to the Grassmannian.

7 Density, factorial examples, and the questions answered

7.1 The counterexample interval

Put $T_k = \binom{k}{2}$. A direct simplification of (1.4) gives

$$\begin{aligned} D_{\text{prime}}(h) &= 2h - 2k + 1 && \text{if } T_k + 2 \leq h \leq T_{k+1}, \\ D_{\text{prime}}(T_{k+1} + 1) &= 2(T_{k+1} + 1) - 2k. \end{aligned} \tag{7.1}$$

These identities follow by bounding the square roots between consecutive integers; in the first interval $(2k-1)^2 < 8h-7 < (2k+1)^2$, with the endpoints of the integer floors giving the displayed value.

For $T_k + 2 \leq h \leq \lfloor k^2/2 \rfloor$, write $c = h - T_k$. Then $2 \leq c \leq \lfloor k/2 \rfloor$, so Theorem 6.1 applies and its invariant is

$$a = k - 2c = k^2 - 2h \geq 0.$$

The variety at $M = D_{\text{prime}}(h)$ is a normal integral complete intersection by [2, Theorems 1 and 2], but its indicated local ring is not rational in characteristic zero and not F -rational in odd characteristic. For strict inequality $h < k^2/2$, the invariant is positive and gives failure of log canonicity and of F -purity.

The interval contains $\lfloor k/2 \rfloor - 1$ integers. Through the completed block ending at $T_{K+1} + 1$, their total number is

$$\sum_{k=4}^K \left(\left\lfloor \frac{k}{2} \right\rfloor - 1 \right) = \left\lfloor \frac{(K-2)^2}{4} \right\rfloor.$$

The endpoint is $K^2/2 + O(K)$; the ratio tends to $1/2$. An incomplete last block changes the count by $O(K)$, proving natural density. The equality cases $a = 0$ occur only once for each even k , so removing them leaves density $1/2$ for the strict failures as well.

7.2 Factoriality

The UFD bound in [2, Theorem 3], for $h \geq 4$, is the minimum of an even candidate and the odd candidate

$$2 \left\lceil \frac{2h - 1 - \sqrt{8h - 31}}{2} \right\rceil + 1. \tag{7.2}$$

For $4 \leq c \leq \lfloor k/2 \rfloor$,

$$8h - 31 = (2k - 1)^2 + 8c - 32, \quad 2k - 1 \leq \sqrt{8h - 31} < 2k + 1.$$

Expression (7.2) is therefore $2h - 2k + 1 = M$, so the frame ring is factorial. Removing $c = 2, 3$ from each block removes only $O(\sqrt{H})$ integers up to H . The factorial subfamily, and its strict non-log-canonical and non- F -pure subfamily, still have natural density $1/2$. This completes the proof of Theorem 1.3.

7.3 Examples and scope

When $a = 0$, write $k = 2c$. Then

$$h = 2c^2, \quad M = (2c - 1)^2, \quad \delta = c. \tag{7.3}$$

The generic transverse projective variety is a smooth intersection of c quadrics in $\mathbb{P}^{2c-1}$ with trivial canonical bundle. For $c = 2$ it is an elliptic quartic curve; for $c = 3$ it is a K3 surface of degree eight; for $c = 4$ it is a Calabi–Yau threefold. For $c \geq 4$, the ambient frame ring is factorial. In odd characteristic these generic transverse cones are F -pure but not F -rational.

M	h	(k, c)	Characteristic-zero conclusion
9	8	(4, 2)	not rational; elliptic transverse cone
15	12	(5, 2)	not log canonical
25	18	(6, 3)	not rational; K3 transverse cone
49	32	(8, 4)	factorial, not rational
63	40	(9, 4)	factorial, not log canonical

The two families answer distinct questions. At the triangular complete-intersection parameters, the principal component has an explicit defining ideal and is normal with a controlled failure of Cohen–Macaulayness. At the prime parameters in (1.5), the full frame variety is already normal and integral, yet its generic transverse cones obstruct rationality. Thus neither normality nor factoriality supplies the proposed rational-singularity conclusion.

Several classification problems remain. The normality of all principal components is not decided by the triangular family, and their defining ideals outside that family remain to be determined. Likewise, $a < 0$ in Theorem 6.1 controls only the generic maximal-isotropic stratum; it gives no assertion about every special point or about other strata. Determining the exact rational-singularity threshold requires this additional local analysis.

Acknowledgment. The author acknowledges OpenAI GPT-6 for assistance with research exploration, proof development, verification code, and manuscript preparation.

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